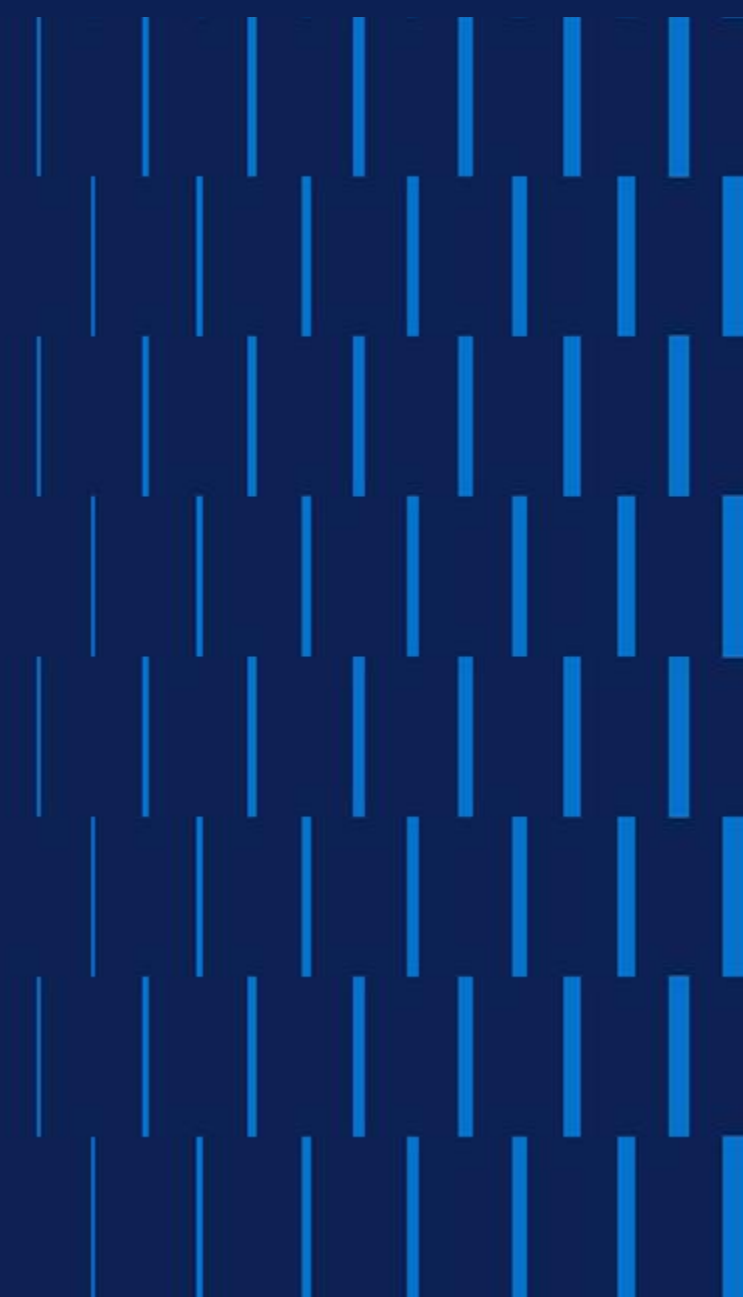


Dell <> OpenQSE



18-May-2026

Smriti Bajaj – Applied Researcher, Quantum Computing

Dell Research Office

Who we are:

A global team of senior technologists, data scientists, customer engagement specialists, and world-class hardware and software engineers and architects.

What we do:

Conduct research and build proof of concepts that influence the technical strategy of Dell Technologies.



Quantum
Computing



AI-Driven
Enterprise



Security



As-a-Service



Immersive
Enterprise

Hybrid Quantum-Classical Computing

No workload runs purely on quantum hardware today. The value is in seamless orchestration across classical HPC, AI accelerators, and quantum backends — a heterogeneous compute fabric.

INITIATIVES

Infrastructure

To develop a comprehensive hardware & software architecture blueprint for hybrid quantum-classical computing systems incorporating accelerator interconnects such as NVQlink

Management & Orchestration

To develop an intelligent orchestration and resource management service for hybrid quantum-classical systems.

Use Case Performance Analytics

To understand where quantum & HQCC create real value for customers by evaluating quantum advantage, economic viability, & readiness so we can guide customers through emerging quantum opportunities.

Quantum & AI

To investigate classical AI enhancement of quantum computing steps and quantum acceleration of ML training, as well as exploring quantum machine learning's potential for Dell and its customers.

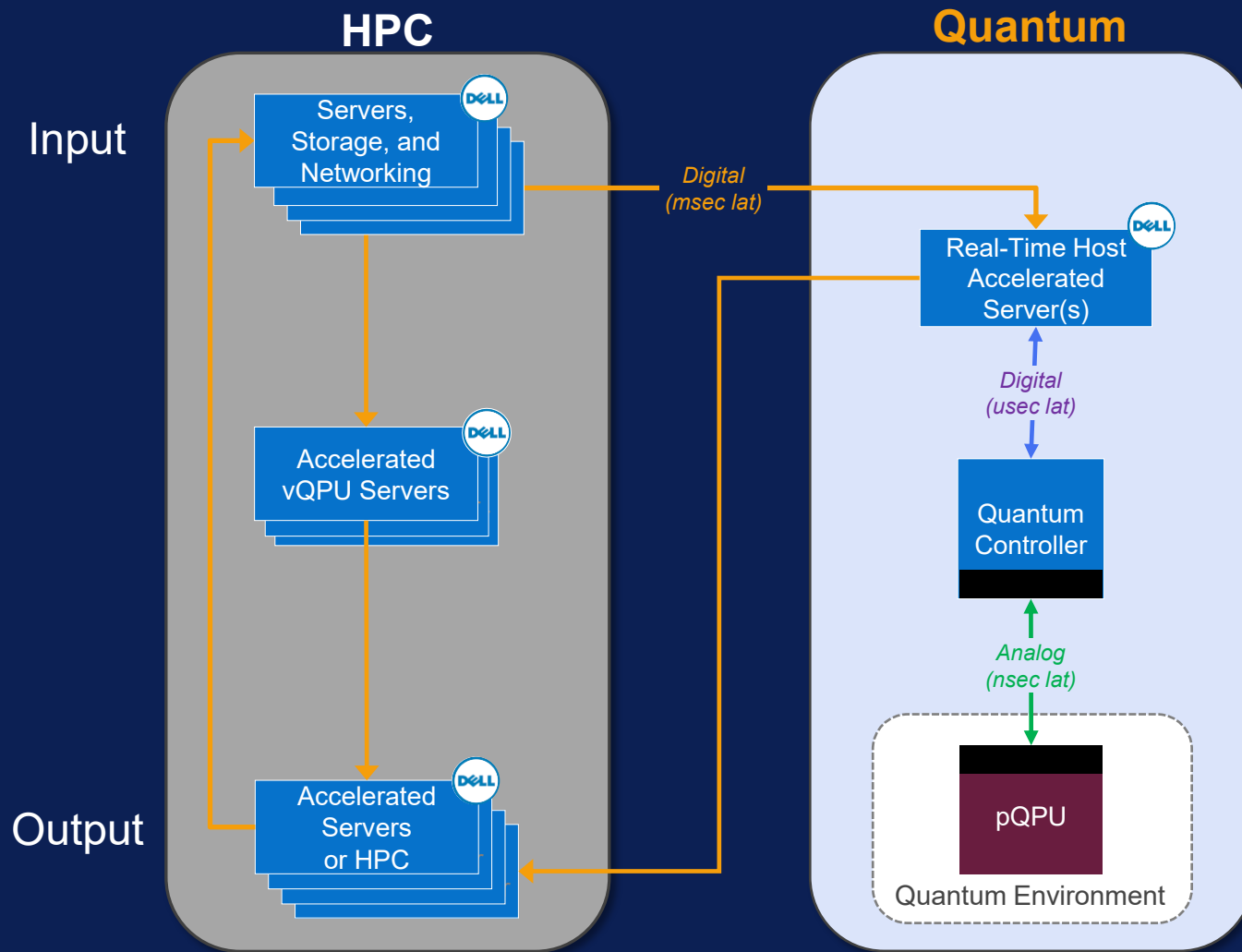
Error Handling

To enhance Dell's understanding of quantum error handling through focused research on classical computation's role applied to error correction, mitigation, and suppression

Quantum on Classical

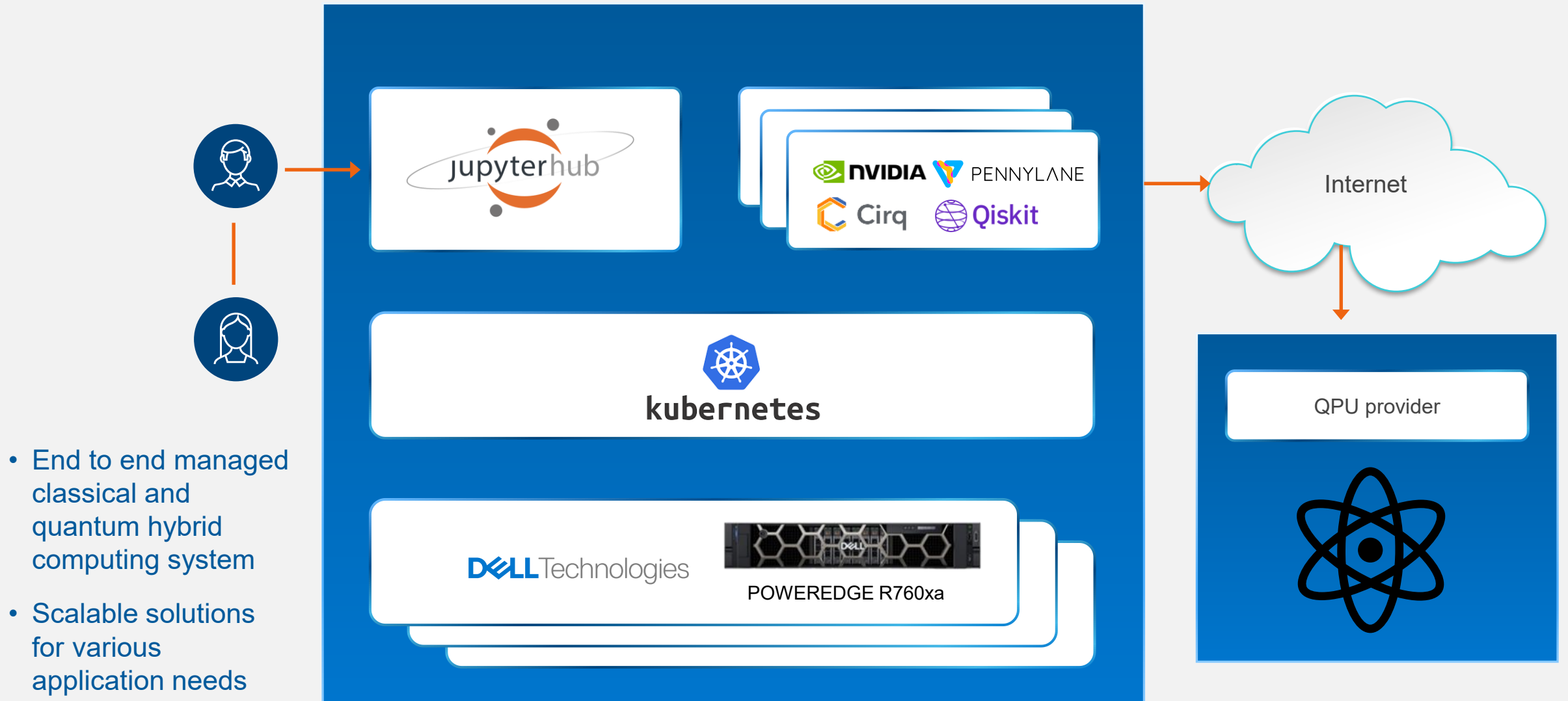
To systematically evaluate dequantized algorithms, quantum-inspired methods, and efficient quantum emulation approaches

Quantum doesn't work without classic infrastructure

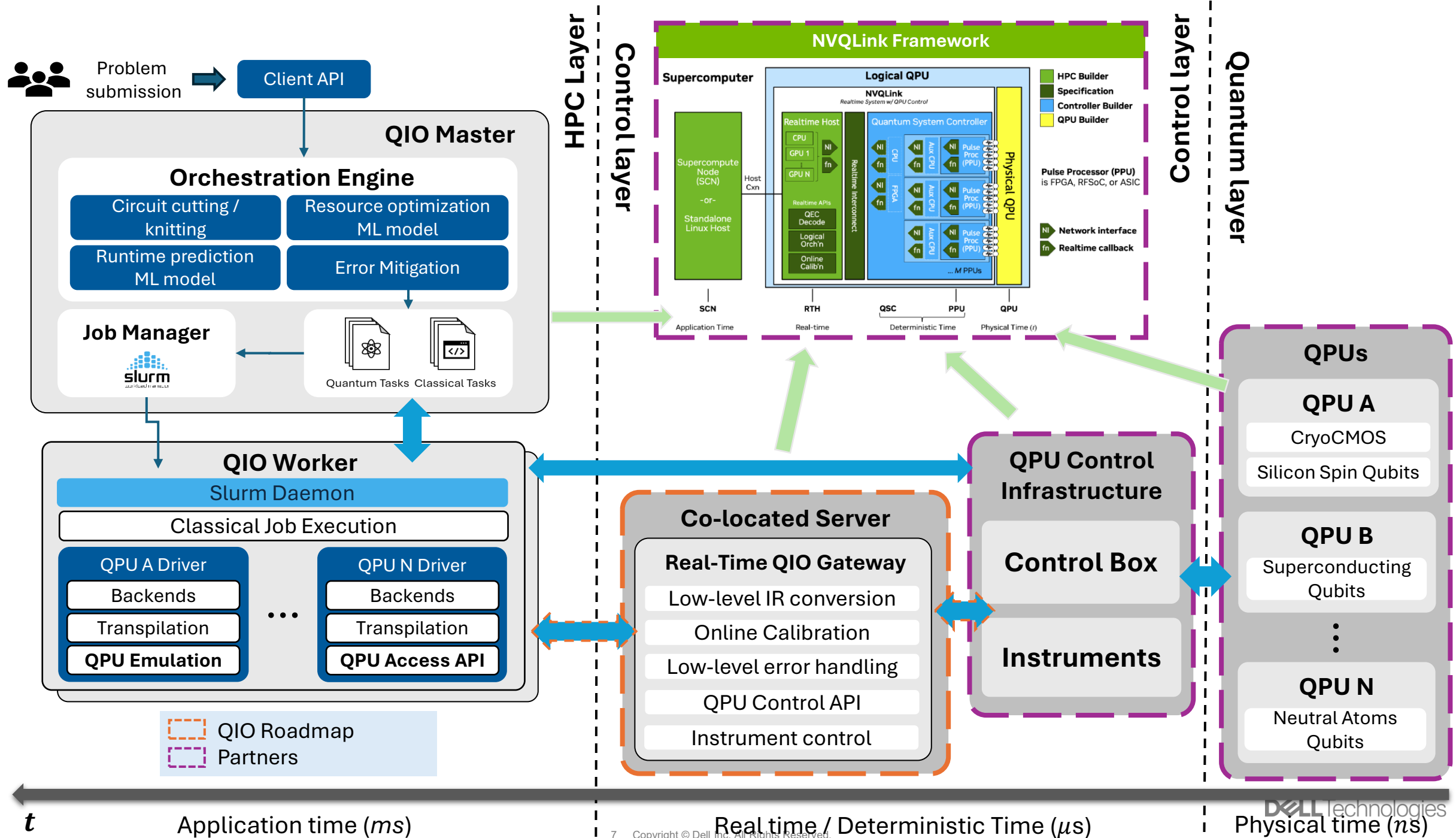


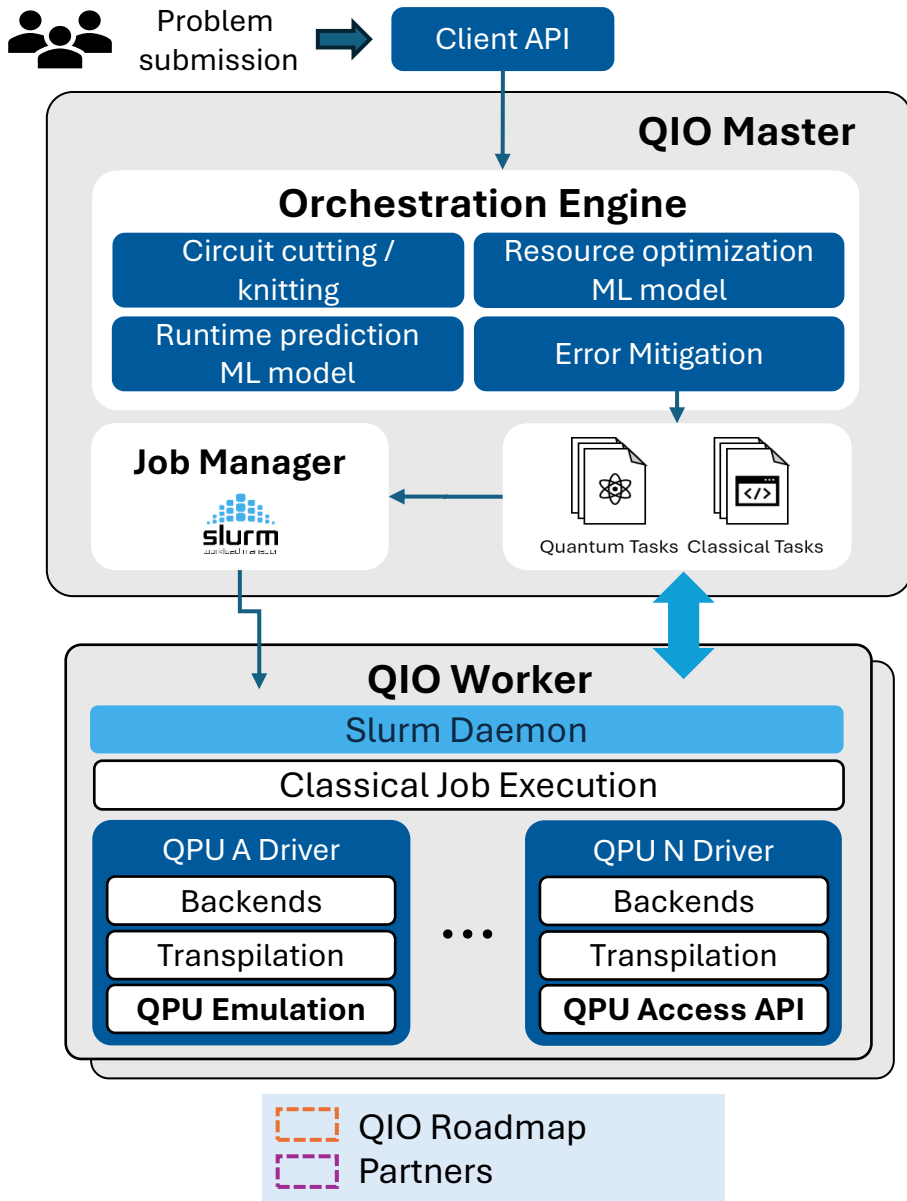
- As the capabilities of QPUs scale up, more classic infrastructure is needed to support it!
- A 'full-stack' that includes emulation (vQPU) and physical QPUs (pQPU)
- An accelerated server functions as a Real-Time Host to field pQPU requests and serve AI-acceleration needs, including for online calibration and QEC (where applicable).
- Multi-Vendor system with location independence

Quantum Intelligent Orchestrator (QIO 1.0)



Current QIO



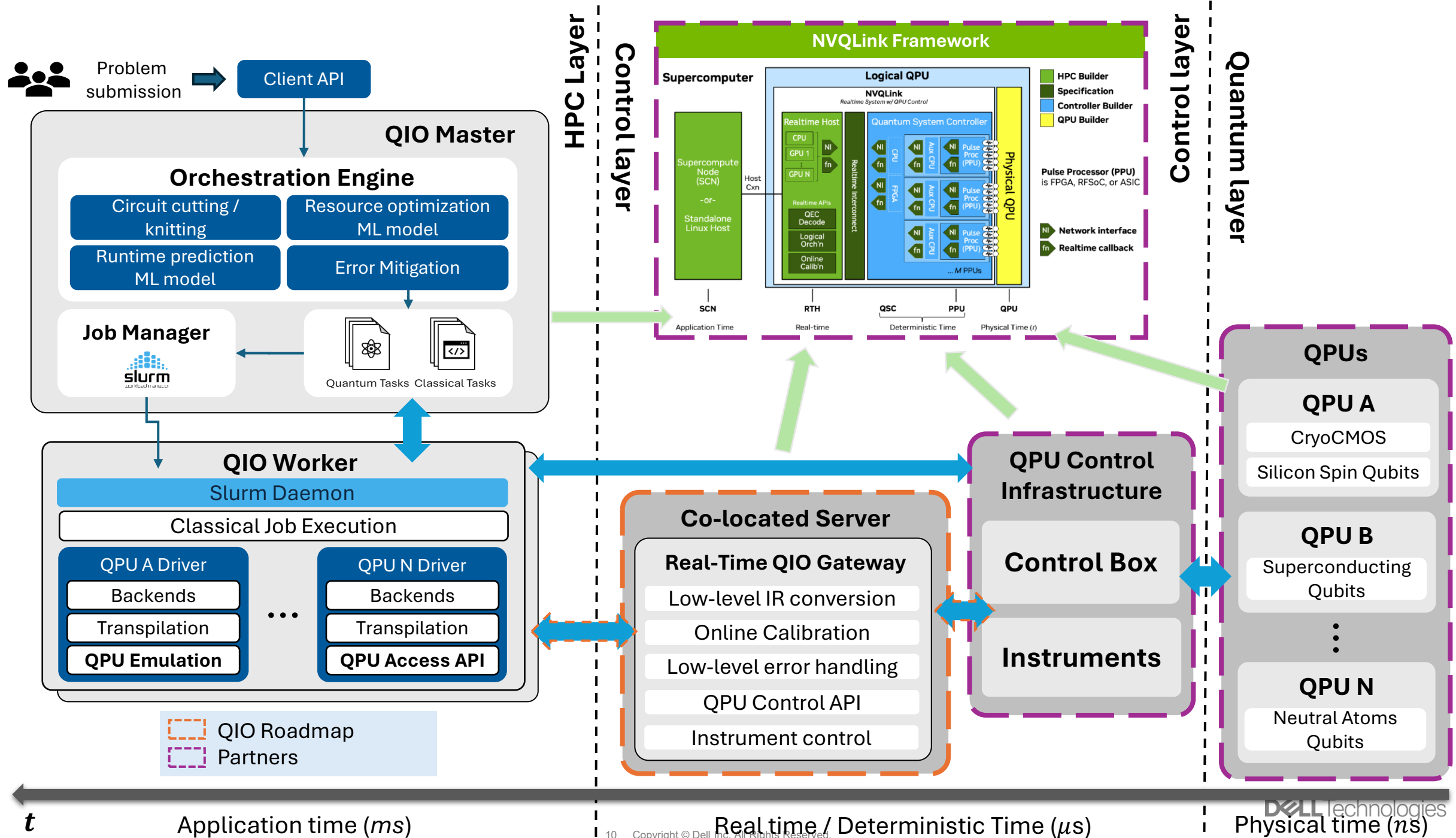


HPC Layer

t

Application time (ms)

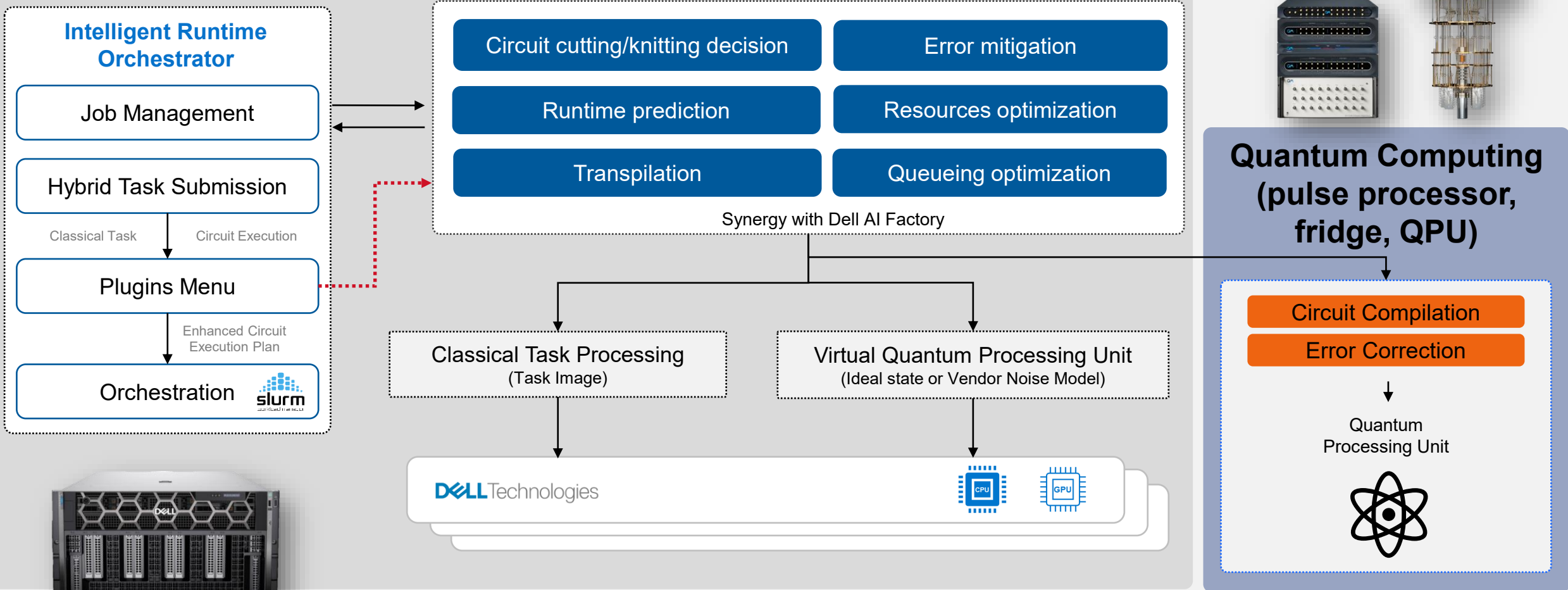




Quantum Intelligent Orchestrator (QIO)

Dell value-add for hybrid workflows

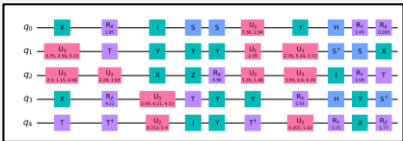
Classical Computing Control Plane for Hybrid Classical Quantum Accelerator



QIO steps on circuit cutting

Example

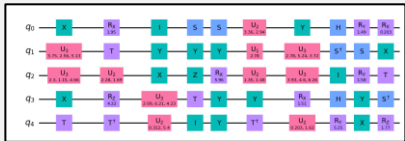
User Input



Circuit



Service-level
Objectives

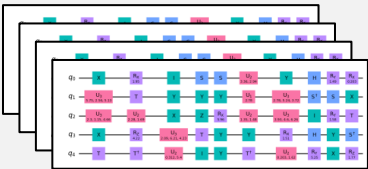


33 Qubit Circuit

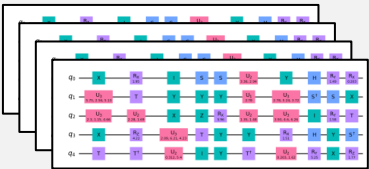


Fastest Execution
Do not exceed \$1000.00

Circuit Cutting



Cutting Circuit in N smaller Sub-Circuits



Cut into 4 Sub-Circuit
Each less than 28 qubits (max qubits)

Runtime
Prediction



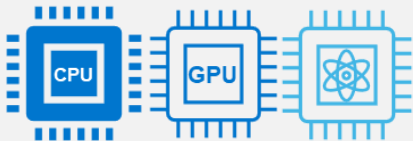
Execution time and Memory Consumption
Prediction when circuit is executed in different
environments

Sub-Circuit 1
Intel
15.69s
19.32GB Ram

Sub-Circuit 4
NVIDIA A100
3s
18.27GB Ram

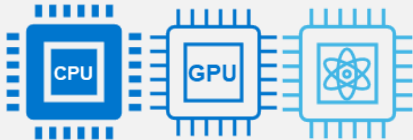
Runtime prediction of each subcircuit for each environment

Resource
Optimization



Determine optimal circuit placement based on
Service-level Objectives and resource availability

Sub-Circuit 1: Intel CPU Simulation
Sub-Circuit 2: NVIDIA A100
Sub-Circuit 3: IonQ QPU
Sub-Circuit 4: IonQ Remote Simulator



Fastest execution time while not exceeding user-defined budget

Deployments with Partners

SC25



QuEra lab

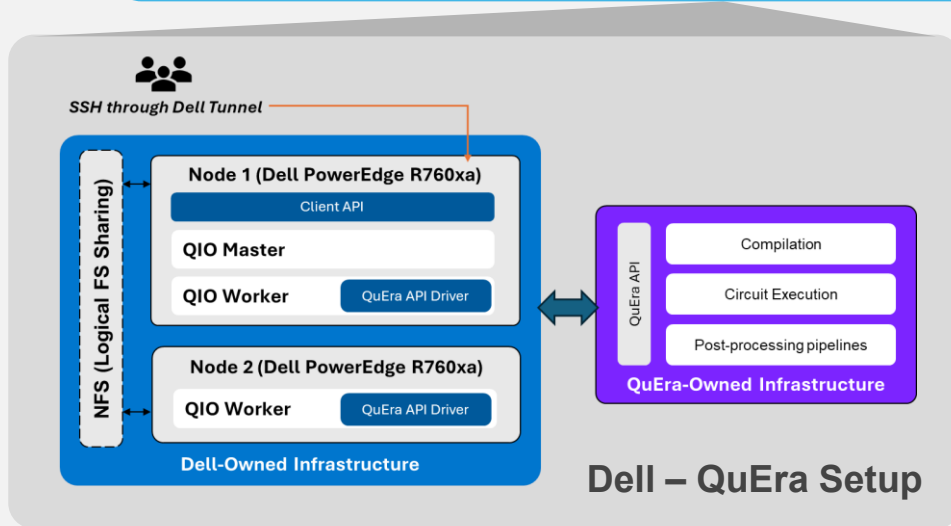
- 1st PoC to connect HPC mini-cluster with QPU side-by-side
- Demo of this environment served as basis for SC'25 presence

DTW26



Equal1

- Integration happening in Cork
- DTW demo planned



University of Washington

- Exploration



Arizona State University

- Joint with Cr8dl



Diraq lab

- Integration happening in Sydney
- Supporting the development of Diraq's roadmap

Home > Dashboards > qlo

 storage (Python 3.10.12)

☆
Edit
Export ▾
Share ▾

<
🕒 Today ▾
>
🔍
🔄 Refresh 5s ▾

Python

[]

make - gio + v [icon] [trash] ... | [full screen] x

```
dqcs-gio | INFO | gio.manager.manager: __init__:31 | manager...
```

Memory usage in GB for the partitions

No data

Tasks timeseries distribution

No data

No data

No data